



Dr. Madani BELACEL

Associate Professor (MCB) — Teacher-Researcher

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ACADEMIC PROFILE

Experienced teacher-researcher at the University of Mostaganem since **2006**, with solid expertise in **language didactics**, **ICT**, **documentary research**, **computer networks / IoT** and **digital pedagogy**. Author of open-source IoT routing protocols (RPL-MQoS, RPL-AER, AER-MQoS, D-SERAN).

Outstanding teaching output: **800+ pages of course materials**, **10,000+ audio resources**, **1,400+ presentations**, academic website of **2,600+ HTML pages**. University habilitation dossier (Option A) prepared.

ACADEMIC EXPERIENCE

University of Mostaganem

Since 2006 — Associate Professor (MCB) / Teacher-Researcher

- Teaching across **6 modules** covering **25+ university levels** (Bachelor → Master / ENS)
- Design of **800+ pages of course materials** and **1,400+ PowerPoint presentations**
- Development of **10,000+ audio resources** (pedagogical dialogues FR/EN)
- Integration of ICT and digital tools into higher education
- Student mentoring and design of programmes meeting university standards
- Creation of a **professional academic website** (GitHub Pages)

TEACHING MODULES

Module	Levels	Language	Resources
ICT	Bachelor → Master	FR/EN	24 lectures + 24 tutorials + 464 MP3
Computer Science ENS	ENS 1st + 2nd year	FR/EN	36 lectures + 36 tutorials + 16 labs
Mostaganem Networks	ISIL-S4 → Master RSID	FR	50+ lectures + 40+ tutorials + 25+ labs + 27 exams
Documentary Research	Master 2 DLE	FR/EN	10 lectures + 10 tutorials
Research & IoT Articles	Research protocols	FR	51 lectures + 61 labs + GitHub
1st-Year English Licence	Licence 1	EN/FR	53 lectures + 5,035 MP3 + 53 PPTX

AREAS OF EXPERTISE

- **Language didactics**: and digital pedagogy
- **ICT and e-learning**: multimedia educational resource design
- **Fundamental computer science**: algorithms, Python, Scratch, OOP, databases, web
- **Networks & IoT**: TCP/IP, RPL, QoS, Contiki-NG, Cooja, Wireshark, security
- **Documentary research**: Zotero, APA 7, research methodology
- **Pedagogical design**: bilingual FR/EN multimedia materials

ACADEMIC PRODUCTION & RESEARCH

Online teaching platform

- Professional website: **2,600+ HTML pages**, 15,500+ files — bilingual FR/EN content
- 7 university levels covered; mock exams with answer keys (27 Network subjects)
- Pedagogical audio dialogues: **10,000+ interactive MP3 files**

IoT / Networks research projects (open-source on GitHub)

- **RPLMQoS BELACEL** — Multi-path QoS-aware RPL extension for IoT networks (Contiki-NG)
- **RPL-AER** — Energy-aware, secure and predictive RPL routing (precision agriculture)
- **AER-MQoS** — Multi-objective extension (QoS, energy, link reliability)
- **D-SERAN** — Self-healing routing protocol for MANETs (trust + energy harvesting)
- **RPL-ClusterIDS** — Distributed hierarchical IDS for RPL-based IoT networks
- **RPLCB** — Source code accompanying a Computer Networks journal paper

TECHNICAL SKILLS

Teaching tools

- Microsoft Office (Word, Excel, PowerPoint)
- Audio / multimedia production & LMS
- Markdown → LaTeX → PDF (pipeline)

Programming languages

- Python, C, HTML/CSS, JavaScript
- Algorithms & structured programming

Research & networking tools

- Zotero, Mendeley — APA 7 standards
- Git / GitHub (version control)
- Contiki-NG, Cooja, Cisco Packet Tracer
- Wireshark, network simulators
- Contiki / RPL / IoT protocol stacks

LANGUAGES & EDUCATION

Language	Level
Arabic	Mother tongue
French	Advanced proficiency (C1)
English	Academic & pedagogical proficiency (B2/C1)

Education

- **PhD in Science** — Specialisation in progress
- **Master's in Computer Science** — Univ. of Mostaganem
- **Bachelor's in English** — Univ. of Mostaganem
- **University Habilitation (HDR)** — Option A dossier prepared

PROFESSIONAL QUALITIES

- Scientific and pedagogical rigour
- Strong organisational and structuring skills
- Didactic innovation and ICT integration
- Commitment to higher-education quality
- Team spirit and student mentoring
- Ability to design coherent and effective resources